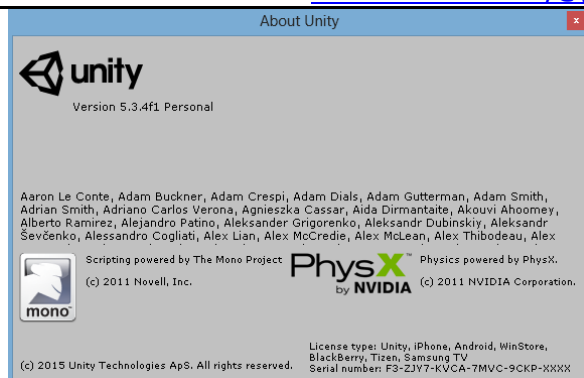


Easy Modular Static Google Maps and GPS

1. GENERAL INFORMATION

DATE OF DOCUMENT	05/08/2016
NAME OF THE PROJECT	Easy Modular Static Google Maps and GPS
AUTHOR	Michael Soler
UNITY VERSION	5.3.4.F1 PERSONAL (NEWER VERSIONS DO NOT WORK)
CONTACT	michael.soler.beatty@gmail.com



Includes video tutorial at the end of the document

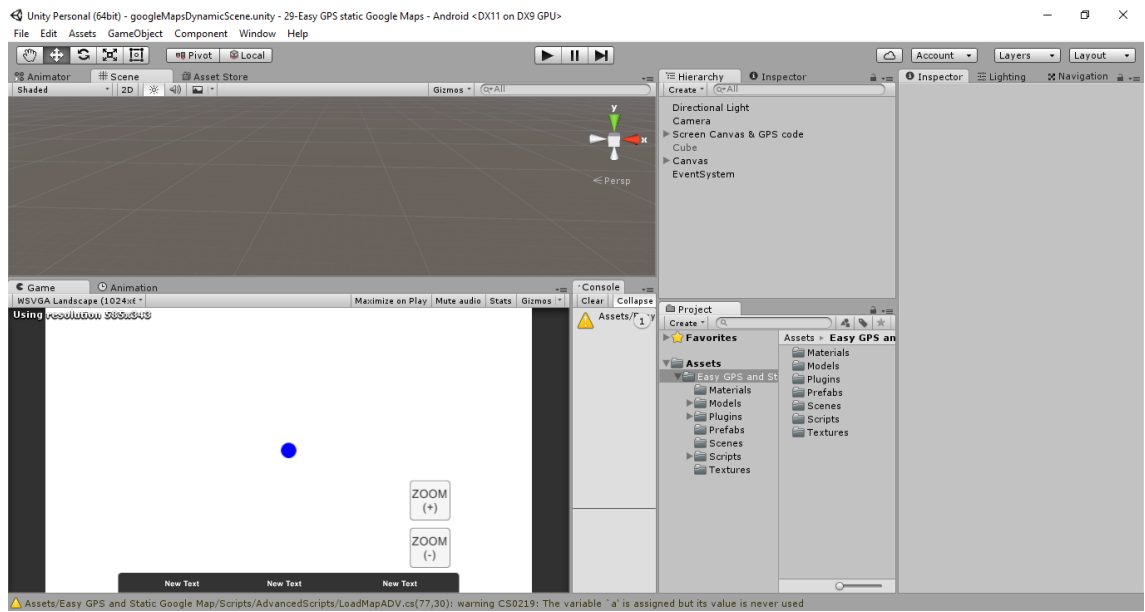


Index

1. GENERAL INFORMATION.....	1
2. IMPORTING INFORMATION	1
3. PROJECT DESCRIPTION	3
4. LAYERS, TAGS AND COLLIDERS.....	4
5. SCRIPTING INFORMATION.....	6
6. VIDEO TUTORIAL FOR ADVANCED TECHNICAL EXPANATIONS	7

2. IMPORTING INFORMATION

This package works independently. Once downloaded you should see something like this:



3. PROJECT DESCRIPTION

This asset gives a starting point to get your GPS working into a Static google maps. The script calls to the google Static API and generates a Texture that is updated during gameplay. The user can change the zoom by script. The idea of this script is to prevent loading “big” images into our device and use a solution of type “mosaic” or “modular” to render google Static Maps.

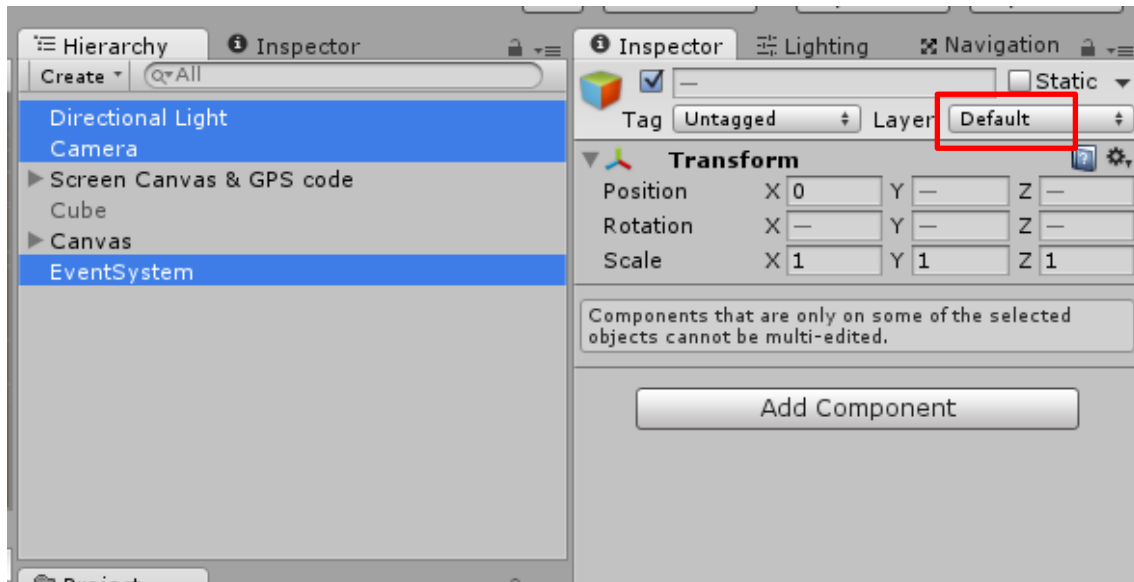
This package contains the following:

- The necessary FBX, textures, models, scripts shown in the video.
- Scripts manage the calls to the GPS and the Google Maps API.
- Two demo scenes: one which uses the GPS data to display the player’s location and where the map is updated after a certain time; a second one, which dynamically creates smaller sectors (portions) of maps, when the cursor is moved manually by drag.
- Complete documentation with video-tutorial to understand the principles of each package and full email support at: michael.soler.beatty@gmail.com. This program was tested on PC and Android.

4. LAYERS, TAGS AND COLLIDERS

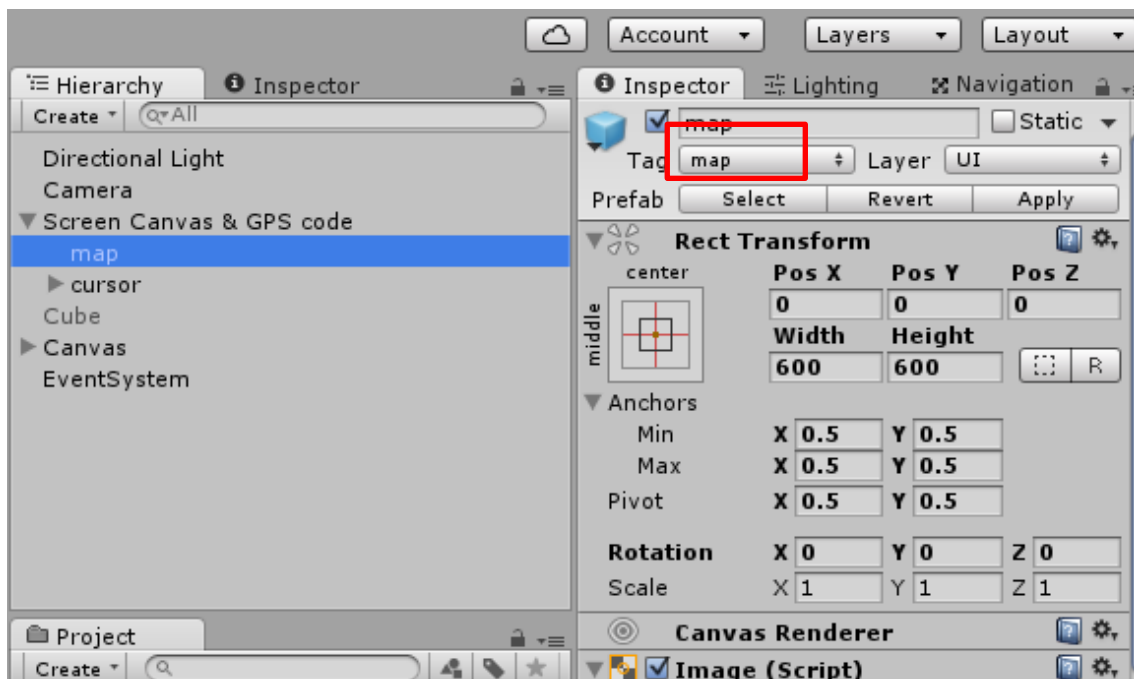
LAYERS

The only layers that are used are the default for regular game-objects, and UI for the GUI objects.



TAGS:

Map sectors are under the tag of “map” while “cursor” it is used to tag the marker.

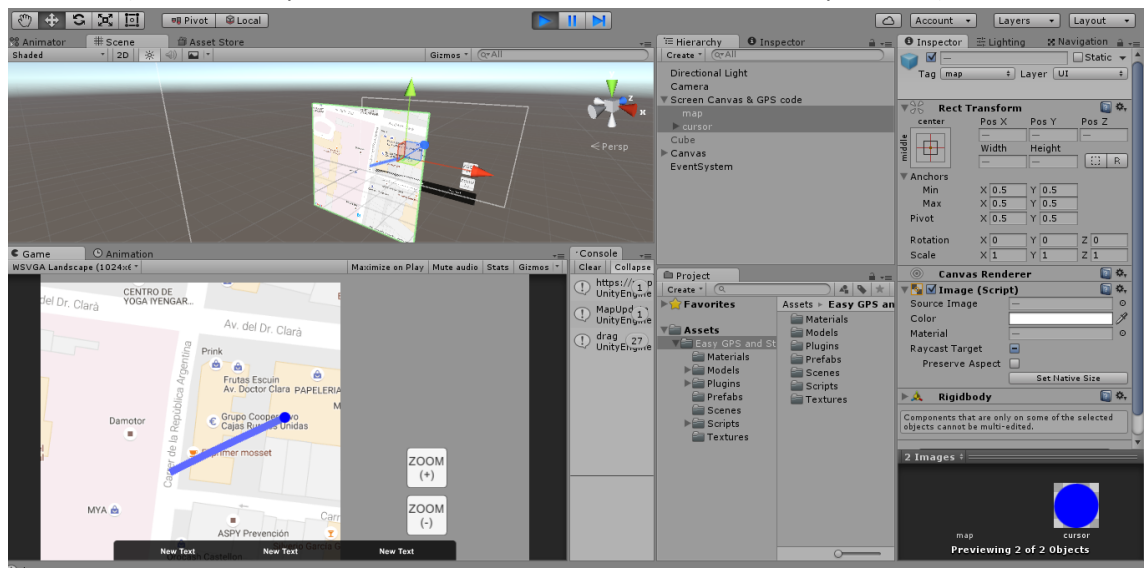


5. SCRIPTING INFORMATION

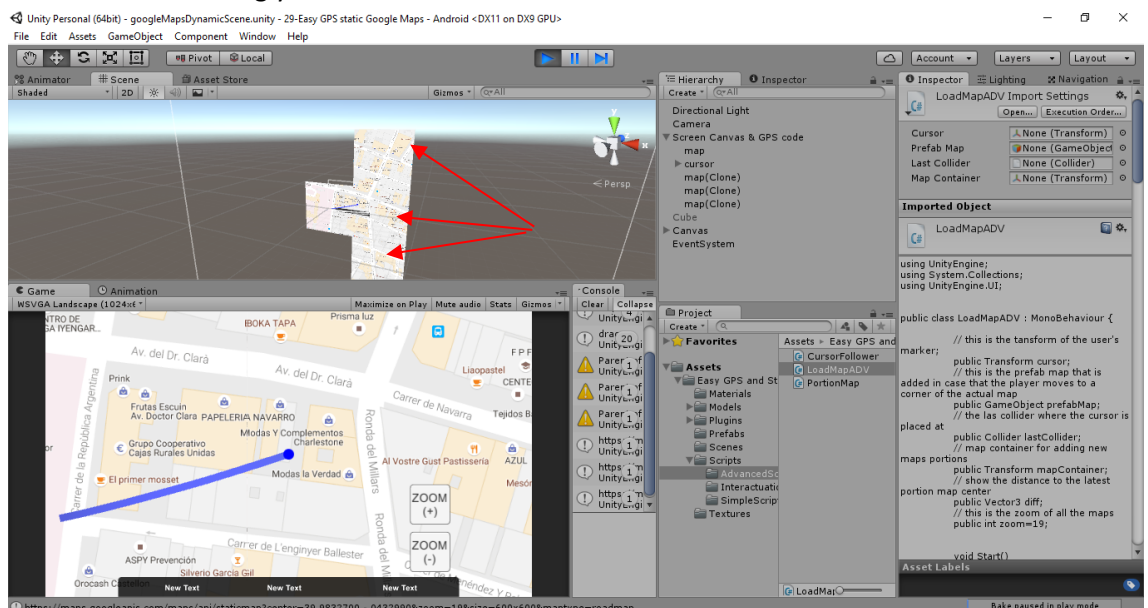
We have divided the scripts into two folders. The first one contains some simple scripts to understand the calls to the Google Maps API and manages the GPS tracking. The second one has more advanced scripts that generate the “mosaic” map.

This is how it is supposed to work:

1. An initial seed of map is created.
2. By pressing and dragging the mouse, the cursor starts to move simulating the GPS feedback (this is not implemented, but can be deduced from the simple scene).

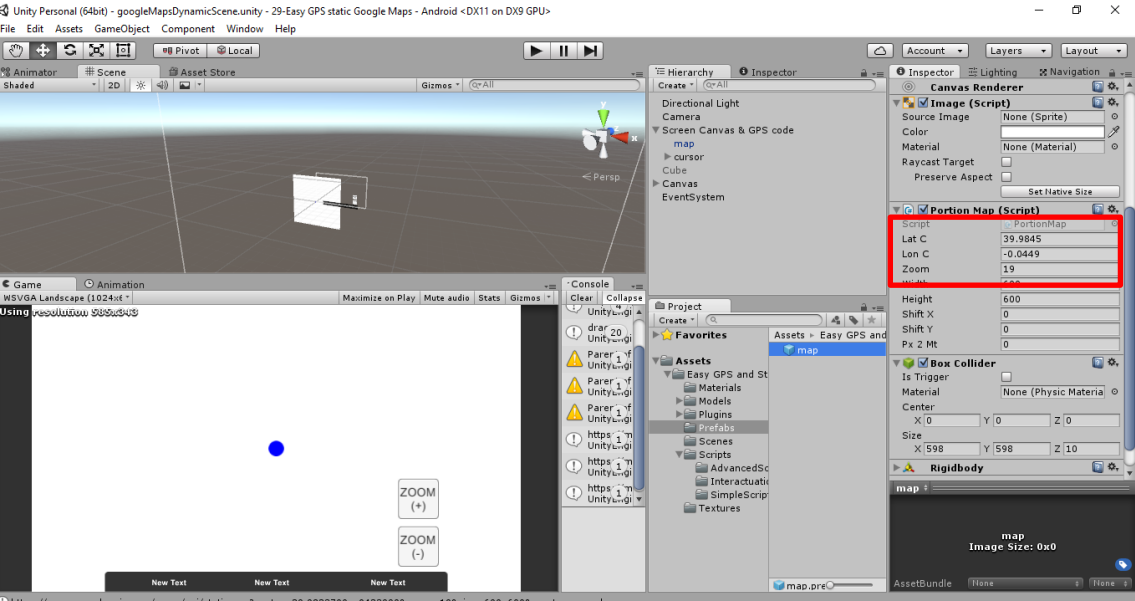


- When the user gets to the limit (when a `collisionExit()` is called), three portions of maps are created accordingly.



4. A map sector is only added if there is nothing at the given position, which means that portions never duplicate.

You can change the default latitude and longitude of the starting map in the prefab:



This is the main scripsts. For future information, please check out the VideoTutorial.

GPSupdate.cs	
Obtains the values from GPS and changes the cursor position accordingly.	
Important variables	Important functions
<pre>// this is the text where the gps messages will show up public Text debugText; // this is the time between GPS updates public float updateGPStime; float elapsed; // this is the radius of the sphere in which the marker will be placed public float R; // variables that the GPS will modify or the developer will test on the editor. These are // the values of latitude and longitude; public float lat, lon; //marker Object that will be placed according ly to the lat and lon: public Transform marker;</pre>	<pre>//the corrutine that is called to obtain the gps values IEnumerator checkGPS()</pre>

6. VIDEO TUTORIAL FOR ADVANCED TECHNICAL EXPLANATIONS

We are trying to give users a more advanced technical service by creating some videos that explain the main aspects of our packages and gives a detailed explanation about how the package works. You will find this tutorial at the following link:



<https://www.youtube.com/watch?v=10ObkTTdu-Q>

7. HOW TO CHANGE MAP PARAMETERS

You can modify a hole bunch of parameters when you want to show he google maps. To add more parameters to you scripts go to the following lines of code in “portionMap.cs”:

You can add your variables here

```
public string callGoogMapApi(float latt, float lonn, int ZM, int wid, int height )
{
    string urlp="https://maps.googleapis.com/maps/api/staticmap?center=";
    urlp+= latt.ToString("##.0000000") + "," + lonn.ToString("##.0000000");
    urlp += "&zoom=" + ZM + "&size=" + width + "x" + height + "&motype=" + "roadmap";

    return urlp;
}
```

You can access all the map's properties at the following link:

<https://developers.google.com/maps/documentation/android-api/?hl=es>